

13 (a) e.g. background count (rate)/radiation

multiple possible counts from each decay

radiation emitted in all directions

dead-time of counter

(daughter) product unstable/also emits radiation

self-absorption of radiation in sample or absorption in air/detector window

three sensible suggestions, 1 each

B3 [3]

(b) $A = A_0 \exp(-\ln 2 \times t / T_{1/2})$

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\ln 2 \times 42.0 / T_{1/2})$$

or

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\lambda \times 42.0)$$

C1

$$T_{1/2} = 5.1 \text{ minutes (306 s)}$$

A1 [2]

(c) discrete energy levels (in nuclei)

B1 [1]